

1. Draw Truth table for JK-FF and generate Excitation table for JK-FF and perform all required steps? [2]

Old State	J	K	Next State
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

~~Old J K Next~~

Excitation Table

Old	New	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

2. Perform Analysis of the circuit A given

[10]

$$Y_A = x \cdot B$$

$$\bar{J}_1 = \bar{x} + Q_0$$

$$J_0 = x + Q_1$$

$$\bar{Y}_B = x$$

$$K_1 = x \cdot Q_1$$

$$K_0 = x$$

$$Z = \bar{x} \cdot Q_1 \cdot Q_0$$

Next States

x	Q ₁	Q ₀	J ₁	K ₁	J ₀	K ₀	Z	Q ₁	Q ₀
0	0	0	1	0	0	0	0	1	0
0	0	1	1	0	0	0	0	1	1
0	1	0	1	0	1	0	0	1	1
0	1	1	1	0	1	0	1	1	1
1	0	0	0	0	1	1	0	0	1
1	0	1	1	0	1	1	0	1	0
1	1	0	0	1	1	1	0	0	1
1	1	1	1	1	1	1	0	0	0

	$\bar{Q}_0\bar{Q}_1$	$\bar{Q}_0Q_1$	$Q_0\bar{Q}_1$	Q_0Q_1
$\bar{x}$	0	0	1	0
x	0	0	0	0

$$x = \bar{x}Q_0Q_1$$

